Module V: Nonlinear Programming Algorithms

Optimization Techniques (24EST372)

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Outline

Objectives

- Understand unconstrained nonlinear programming algorithms.
- Learn properties of unimodal single-variable functions and interval of uncertainty.
- Master direct search methods: dichotomous and golden section.
- Apply gradient methods to twice continuously differentiable functions.
- Practice through exercises and assignments, focusing on AI and computer science applications.

Unconstrained Optimization

Optimize a nonlinear function without constraints:

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Goal: Find $\mathbf{x}^*$ that optimizes $f(\mathbf{x})$ using iterative algorithms, leveraging gradient or direct search methods.

Unimodal Single-Variable Functions

A function $f(x)$ is unimodal if it has a single extremum (minimum or maximum) in an interval $[a, b]$.

For minimization: $f(x^*) \leq f(x)$ for all $x \in [a, b]$, and $f(x)$ is strictly increasing for $x > x^*$, decreasing for $x < x^*$.

Enables efficient search by narrowing the interval containing the extremum.

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Example: $f(x) = x^2$ (unimodal, minimum at $x = 0$).

Interval of Uncertainty

The interval $[a, b]$ containing the optimum x^*.

Iterative methods reduce $[a, b]$ until $|b - a| < \epsilon$, where ϵ is the tolerance.

Direct search methods (e.g., dichotomous, golden section) systematically shrink this interval.

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Goal: Achieve high precision with minimal function evaluations.

Dichotomous Search

For a unimodal function $f(x)$ on $[a, b]$:

- Choose two points $x_1 = \frac{a+b}{2} - \delta$, $x_2 = \frac{a+b}{2} + \delta$, where δ is small.
- Evaluate $f(x_1)$, $f(x_2)$.
- If $f(x_1) < f(x_2)$, new interval is $[a, x_2]$; else, $[x_1, b]$.

Repeat until interval length $|b - a| < \epsilon$.

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If $f(x_1) < f(x_2)$, new interval is $[a, x_2]$; else, $[x_1, b]$.

Repeat until interval length $|b - a| < \epsilon$.

Advantage: Simple, requires only function evaluations.

Golden Section Search

Uses the golden ratio $\phi = \frac{1+\sqrt{5}}{2} \approx 1.618$ to divide $[a, b]$.

Choose points $x_1 = b - \frac{b-a}{\phi}$, $x_2 = a + \frac{b-a}{\phi}$.

Evaluate $f(x_1)$, $f(x_2)$:

If $f(x_1) < f(x_2)$, new interval is $[a, x_2]$.
Else, new interval is $[x_1, b]$.

Reuse one point to minimize evaluations.

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Reuse one point to minimize evaluations.

Advantage: Efficient, converges faster than dichotomous search.

Gradient Method (Steepest Ascent)

For twice continuously differentiable $f(\mathbf{x})$, maximize by iterating:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \nabla f(\mathbf{x}_k)$$

α_k: Step size (via line search or fixed).

Direction $\nabla f(\mathbf{x}_k)$: Steepest ascent for maximization.

For minimization, use steepest descent: $\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \nabla f(\mathbf{x}_k)$.

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Applications: Neural network training, parameter optimization.

Step-by-Step Procedure for Unconstrained Optimization

To optimize $f(\mathbf{x})$:

Determine Function Type: Check if f is unimodal (single-variable) or twice differentiable (multivariable).

Choose Algorithm:

- Single-variable: Dichotomous or Golden Section search.
- Multivariable: Gradient Method (steepest ascent/descent).

Initialize: Select $\mathbf{x}_0$, interval $[a, b]$, or tolerance ϵ.

Iterate:

$$\begin{cases} \text{Dichotomous: Evaluate } f \text{ at two points, update interval.} \\ \text{Golden Section: Use golden ratio to update interval.} \\ \text{Gradient Method: Update } \mathbf{x}_{k+1} = \mathbf{x}_k \pm \alpha_k \nabla f(\mathbf{x}_k). \end{cases}$$

Check Convergence: Stop if interval $|b - a| < \epsilon$ or $\|\nabla f(\mathbf{x}_k)\| < \epsilon$.

Verify Optimality: Check objective value or second-order conditions.

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$$x_1 = 1.5 - 0.1 = 1.4, \quad f(1.4) = (1.4-1)^2 = 0.16$$
$$x_2 = 1.5 + 0.1 = 1.6, \quad f(1.6) = (1.6-1)^2 = 0.36$$

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Since $f(1.4) < f(1.6)$, new interval $[0, 1.6]$.

Solved Problem 1: Solution (Continued)

Step 3: Iteration 2

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Step 3: Iteration 2 Midpoint $m = \frac{0+1.6}{2} = 0.8$.

$$x_1 = 0.8 - 0.1 = 0.7, \quad f(0.7) = (0.7-1)^2 = 0.09$$

$$x_2 = 0.8 + 0.1 = 0.9, \quad f(0.9) = (0.9-1)^2 = 0.01$$

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$$x_1 = 1.15 - 0.1 = 1.05, \quad f(1.05) = (1.05-1)^2 = 0.0025$$

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$$x_1 = 1.5 - 0.1 = 1.4, \quad f(1.4) = 1.4^2 - 2 \cdot 1.4 = 1.96 - 2.8 = -0.84$$

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Minimize $f(x) = e^x - x^2$ on $[0, 2]$ using Dichotomous Search (approximate values).

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$$x_1 = 1 - 0.05 = 0.95, \quad f(0.95) \approx e^{0.95} - 0.95^2 \approx 2.585 - 0.9025 \approx 1.6825$$

$$x_2 = 1 + 0.05 = 1.05, \quad f(1.05) \approx e^{1.05} - 1.05^2 \approx 2.857 - 1.1025 \approx 1.7545$$

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Since $f(0.95) < f(1.05)$, new interval $[0, 1.05]$.

Solved Problem 3: Solution (Continued)

Step 3: Iteration 2

Solved Problem 3: Solution (Continued)

Step 3: Iteration 2 Midpoint $m = \frac{0+1.05}{2} = 0.525$.

$$x_1 = 0.525 - 0.05 = 0.475, \quad f(0.475) \approx e^{0.475} - 0.475^2 \approx 1.609 - 0.2256 \approx 1.3834$$

$$x_2 = 0.525 + 0.05 = 0.575, \quad f(0.575) \approx e^{0.575} - 0.575^2 \approx 1.777 - 0.3306 \approx 1.4464$$

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Step 3: Iteration 2 Midpoint $m = \frac{0+1.05}{2} = 0.525$.

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$$x_2 = 0.525 + 0.05 = 0.575, \quad f(0.575) \approx e^{0.575} - 0.575^2 \approx 1.777 - 0.3306 \approx 1.4464$$

Since $f(0.475) < f(0.575)$, new interval $[0, 0.575]$.

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Step 3: Iteration 2 Midpoint $m = \frac{0+1.05}{2} = 0.525$.

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$$x_2 = 0.525 + 0.05 = 0.575, \quad f(0.575) \approx e^{0.575} - 0.575^2 \approx 1.777 - 0.3306 \approx 1.4464$$

Since $f(0.475) < f(0.575)$, new interval $[0, 0.575]$. **Step 4: Iteration 3**

Solved Problem 3: Solution (Continued)

Step 3: Iteration 2 Midpoint $m = \frac{0+1.05}{2} = 0.525$.

$$x_1 = 0.525 - 0.05 = 0.475, \quad f(0.475) \approx e^{0.475} - 0.475^2 \approx 1.609 - 0.2256 \approx 1.3834$$

$$x_2 = 0.525 + 0.05 = 0.575, \quad f(0.575) \approx e^{0.575} - 0.575^2 \approx 1.777 - 0.3306 \approx 1.4464$$

Since $f(0.475) < f(0.575)$, new interval $[0, 0.575]$. **Step 4: Iteration 3** Midpoint $m = \frac{0+0.575}{2} = 0.2875$.

$$x_1 = 0.2875 - 0.05 = 0.2375, \quad f(0.2375) \approx e^{0.2375} - 0.2375^2 \approx 1.268 - 0.0564 \approx 1.2116$$

$$x_2 = 0.2875 + 0.05 = 0.3375, \quad f(0.3375) \approx e^{0.3375} - 0.3375^2 \approx 1.401 - 0.1139 \approx 1.2871$$

Solved Problem 3: Solution (Continued)

Step 3: Iteration 2 Midpoint $m = \frac{0+1.05}{2} = 0.525$.

$$x_1 = 0.525 - 0.05 = 0.475, \quad f(0.475) \approx e^{0.475} - 0.475^2 \approx 1.609 - 0.2256 \approx 1.3834$$

$$x_2 = 0.525 + 0.05 = 0.575, \quad f(0.575) \approx e^{0.575} - 0.575^2 \approx 1.777 - 0.3306 \approx 1.4464$$

Since $f(0.475) < f(0.575)$, new interval $[0, 0.575]$. **Step 4: Iteration 3** Midpoint $m = \frac{0+0.575}{2} = 0.2875$.

$$x_1 = 0.2875 - 0.05 = 0.2375, \quad f(0.2375) \approx e^{0.2375} - 0.2375^2 \approx 1.268 - 0.0564 \approx 1.2116$$

$$x_2 = 0.2875 + 0.05 = 0.3375, \quad f(0.3375) \approx e^{0.3375} - 0.3375^2 \approx 1.401 - 0.1139 \approx 1.2871$$

Since $f(0.2375) < f(0.3375)$, new interval $[0, 0.3375]$.

Solved Problem 3: Solution (Continued)

Step 3: Iteration 2 Midpoint $m = \frac{0+1.05}{2} = 0.525$.

$$x_1 = 0.525 - 0.05 = 0.475, \quad f(0.475) \approx e^{0.475} - 0.475^2 \approx 1.609 - 0.2256 \approx 1.3834$$

$$x_2 = 0.525 + 0.05 = 0.575, \quad f(0.575) \approx e^{0.575} - 0.575^2 \approx 1.777 - 0.3306 \approx 1.4464$$

Since $f(0.475) < f(0.575)$, new interval $[0, 0.575]$. **Step 4: Iteration 3** Midpoint $m = \frac{0+0.575}{2} = 0.2875$.

$$x_1 = 0.2875 - 0.05 = 0.2375, \quad f(0.2375) \approx e^{0.2375} - 0.2375^2 \approx 1.268 - 0.0564 \approx 1.2116$$

$$x_2 = 0.2875 + 0.05 = 0.3375, \quad f(0.3375) \approx e^{0.3375} - 0.3375^2 \approx 1.401 - 0.1139 \approx 1.2871$$

Since $f(0.2375) < f(0.3375)$, new interval $[0, 0.3375]$. **Convergence**: Approximate minimum near $x^* \approx 0$, $f(0) = 1$.

Solved Problem 3: Solution (Continued)

Step 3: Iteration 2 Midpoint $m = \frac{0+1.05}{2} = 0.525$.

$$x_1 = 0.525 - 0.05 = 0.475, \quad f(0.475) \approx e^{0.475} - 0.475^2 \approx 1.609 - 0.2256 \approx 1.3834$$

$$x_2 = 0.525 + 0.05 = 0.575, \quad f(0.575) \approx e^{0.575} - 0.575^2 \approx 1.777 - 0.3306 \approx 1.4464$$

Since $f(0.475) < f(0.575)$, new interval $[0, 0.575]$. **Step 4: Iteration 3** Midpoint $m = \frac{0+0.575}{2} = 0.2875$.

$$x_1 = 0.2875 - 0.05 = 0.2375, \quad f(0.2375) \approx e^{0.2375} - 0.2375^2 \approx 1.268 - 0.0564 \approx 1.2116$$

$$x_2 = 0.2875 + 0.05 = 0.3375, \quad f(0.3375) \approx e^{0.3375} - 0.3375^2 \approx 1.401 - 0.1139 \approx 1.2871$$

Since $f(0.2375) < f(0.3375)$, new interval $[0, 0.3375]$. **Convergence**: Approximate minimum near $x^* \approx 0$, $f(0) = 1$. **Verification**: Numerical minimum around boundary.

Solved Problem 1: Golden Section Search

Minimize $f(x) = x^2 + 2x + 2$ on $[0, 4]$ using Golden Section Search.

Solved Problem 1: Golden Section Search

Minimize $f(x) = x^2 + 2x + 2$ on $[0, 4]$ using Golden Section Search. **Step 1: Initialize**

Solved Problem 1: Golden Section Search

Minimize $f(x) = x^2 + 2x + 2$ on $[0, 4]$ using Golden Section Search. **Step 1: Initialize** Interval $[a, b] = [0, 4]$, $\phi = \frac{1+\sqrt{5}}{2} \approx 1.618$, tolerance $\epsilon = 0.5$.

Solved Problem 1: Golden Section Search

Minimize $f(x) = x^2 + 2x + 2$ on $[0, 4]$ using Golden Section Search. **Step 1: Initialize** Interval $[a, b] = [0, 4]$, $\phi = \frac{1+\sqrt{5}}{2} \approx 1.618$, tolerance $\epsilon = 0.5$. **Step 2: Compute Points**

Solved Problem 1: Golden Section Search

Minimize $f(x) = x^2 + 2x + 2$ on $[0, 4]$ using Golden Section Search. **Step 1: Initialize** Interval $[a, b] = [0, 4]$, $\phi = \frac{1+\sqrt{5}}{2} \approx 1.618$, tolerance $\epsilon = 0.5$. **Step 2: Compute Points**

$$x_1 = b - \frac{b-a}{\phi} = 4 - \frac{4-0}{1.618} \approx 4 - 2.47 = 1.53$$

$$x_2 = a + \frac{b-a}{\phi} = 0 + \frac{4}{1.618} \approx 2.47$$

Solved Problem 1: Golden Section Search

Minimize $f(x) = x^2 + 2x + 2$ on $[0, 4]$ using Golden Section Search. **Step 1: Initialize** Interval $[a, b] = [0, 4]$, $\phi = \frac{1+\sqrt{5}}{2} \approx 1.618$, tolerance $\epsilon = 0.5$. **Step 2: Compute Points**

$$x_1 = b - \frac{b-a}{\phi} = 4 - \frac{4-0}{1.618} \approx 4 - 2.47 = 1.53$$

$$x_2 = a + \frac{b-a}{\phi} = 0 + \frac{4}{1.618} \approx 2.47$$

Step 3: Evaluate

Solved Problem 1: Golden Section Search

Minimize $f(x) = x^2 + 2x + 2$ on $[0, 4]$ using Golden Section Search. **Step 1: Initialize** Interval $[a, b] = [0, 4]$, $\phi = \frac{1+\sqrt{5}}{2} \approx 1.618$, tolerance $\epsilon = 0.5$. **Step 2: Compute Points**

$$x_1 = b - \frac{b-a}{\phi} = 4 - \frac{4-0}{1.618} \approx 4 - 2.47 = 1.53$$

$$x_2 = a + \frac{b-a}{\phi} = 0 + \frac{4}{1.618} \approx 2.47$$

Step 3: Evaluate

$$f(x_1) = f(1.53) = 1.53^2 + 2 \cdot 1.53 + 2 \approx 7.41$$

$$f(x_2) = f(2.47) = 2.47^2 + 2 \cdot 2.47 + 2 \approx 12.99$$

Solved Problem 1: Solution (Continued)

Step 4: Update Interval

Solved Problem 1: Solution (Continued)

Step 4: Update Interval Since $f(1.53) < f(2.47)$, new interval is $[a, x_2] = [0, 2.47]$.

Solved Problem 1: Solution (Continued)

Step 4: Update Interval Since $f(1.53) < f(2.47)$, new interval is $[a, x_2] = [0, 2.47]$.
Iteration 2:

Solved Problem 1: Solution (Continued)

Step 4: Update Interval Since $f(1.53) < f(2.47)$, new interval is $[a, x_2] = [0, 2.47]$.
Iteration 2:

$$x_1 = 2.47 - \frac{2.47 - 0}{1.618} \approx 2.47 - 1.53 = 0.94$$

$$x_2 = 0 + \frac{2.47}{1.618} \approx 1.53 \text{ (reuse from previous)}$$

$$f(0.94) = 0.94^2 + 2 \cdot 0.94 + 2 \approx 4.76$$

Solved Problem 1: Solution (Continued)

Step 4: Update Interval Since $f(1.53) < f(2.47)$, new interval is $[a, x_2] = [0, 2.47]$.
Iteration 2:

$$x_1 = 2.47 - \frac{2.47 - 0}{1.618} \approx 2.47 - 1.53 = 0.94$$

$$x_2 = 0 + \frac{2.47}{1.618} \approx 1.53 \text{ (reuse from previous)}$$

$$f(0.94) = 0.94^2 + 2 \cdot 0.94 + 2 \approx 4.76$$

Convergence: Interval length $2.47 - 0 = 2.47 > 0.5$. Continue iterations until $|b - a| < 0.5$.

Solved Problem 1: Solution (Continued)

Step 4: Update Interval Since $f(1.53) < f(2.47)$, new interval is $[a, x_2] = [0, 2.47]$.
Iteration 2:

$$x_1 = 2.47 - \frac{2.47 - 0}{1.618} \approx 2.47 - 1.53 = 0.94$$

$$x_2 = 0 + \frac{2.47}{1.618} \approx 1.53 \text{ (reuse from previous)}$$

$$f(0.94) = 0.94^2 + 2 \cdot 0.94 + 2 \approx 4.76$$

Convergence: Interval length $2.47 - 0 = 2.47 > 0.5$. Continue iterations until $|b - a| < 0.5$. **Solution**: Approximate minimum at $x^* \approx 1$, $f(1) = 1 + 2 + 2 = 5$.

Solved Problem 1: Solution (Continued)

Step 4: Update Interval Since $f(1.53) < f(2.47)$, new interval is $[a, x_2] = [0, 2.47]$.
Iteration 2:

$$x_1 = 2.47 - \frac{2.47 - 0}{1.618} \approx 2.47 - 1.53 = 0.94$$

$$x_2 = 0 + \frac{2.47}{1.618} \approx 1.53 \text{ (reuse from previous)}$$

$$f(0.94) = 0.94^2 + 2 \cdot 0.94 + 2 \approx 4.76$$

Convergence: Interval length $2.47 - 0 = 2.47 > 0.5$. Continue iterations until $|b - a| < 0.5$. **Solution**: Approximate minimum at $x^* \approx 1$, $f(1) = 1 + 2 + 2 = 5$. **Verification**: $f'(x) = 2x + 2 = 0 \implies x = -1$ (outside $[0, 4]$), so minimum on $[0, 4]$ is at boundary or interior point close to 1.

Solved Problem 2: Golden Section Search

Minimize $f(x) = (x-2)^2 + 1$ on $[0,3]$ using Golden Section Search.

Solved Problem 2: Golden Section Search

Minimize $f(x) = (x-2)^2 + 1$ on $[0, 3]$ using Golden Section Search. **Step 1: Initialize**

Solved Problem 2: Golden Section Search

Minimize $f(x) = (x-2)^2 + 1$ on $[0, 3]$ using Golden Section Search. **Step 1: Initialize** Interval $[a, b] = [0, 3]$, $\phi \approx 1.618$, tolerance $\epsilon = 0.5$.

Solved Problem 2: Golden Section Search

Minimize $f(x) = (x-2)^2 + 1$ on $[0, 3]$ using Golden Section Search. **Step 1: Initialize** Interval $[a, b] = [0, 3]$, $\phi \approx 1.618$, tolerance $\epsilon = 0.5$. **Step 2: Iteration 1**

Solved Problem 2: Golden Section Search

Minimize $f(x) = (x-2)^2 + 1$ on $[0, 3]$ using Golden Section Search. **Step 1: Initialize** Interval $[a, b] = [0, 3]$, $\phi \approx 1.618$, tolerance $\epsilon = 0.5$. **Step 2: Iteration 1**

$$x_1 = 3 - \frac{3-0}{\phi} \approx 3 - 1.85 = 1.15, \quad f(1.15) \approx (1.15-2)^2 + 1 \approx 0.7225 + 1 = 1.7225$$

$$x_2 = 0 + \frac{3}{\phi} \approx 1.85, \quad f(1.85) \approx (1.85-2)^2 + 1 \approx 0.0225 + 1 = 1.0225$$

Solved Problem 2: Golden Section Search

Minimize $f(x) = (x-2)^2 + 1$ on $[0, 3]$ using Golden Section Search. **Step 1: Initialize** Interval $[a, b] = [0, 3]$, $\phi \approx 1.618$, tolerance $\epsilon = 0.5$. **Step 2: Iteration 1**

$$x_1 = 3 - \frac{3-0}{\phi} \approx 3 - 1.85 = 1.15, \quad f(1.15) \approx (1.15-2)^2 + 1 \approx 0.7225 + 1 = 1.7225$$

$$x_2 = 0 + \frac{3}{\phi} \approx 1.85, \quad f(1.85) \approx (1.85-2)^2 + 1 \approx 0.0225 + 1 = 1.0225$$

Since $f(1.15) > f(1.85)$, new interval $[1.15, 3]$.

Solved Problem 2: Solution (Continued)

Step 3: Iteration 2

Solved Problem 2: Solution (Continued)

Step 3: Iteration 2

$$x_1 = 3 - \frac{3 - 1.15}{\phi} \approx 3 - 1.14 = 1.86, \quad f(1.86) \approx (1.86 - 2)^2 + 1 \approx 0.0196 + 1 = 1.0196$$

$$x_2 = 1.15 + \frac{3 - 1.15}{\phi} \approx 1.15 + 1.14 = 2.29, \quad f(2.29) \approx (2.29 - 2)^2 + 1 \approx 0.0841 + 1 = 1.0841$$

Solved Problem 2: Solution (Continued)

Step 3: Iteration 2

$$x_1 = 3 - \frac{3 - 1.15}{\phi} \approx 3 - 1.14 = 1.86, \quad f(1.86) \approx (1.86 - 2)^2 + 1 \approx 0.0196 + 1 = 1.0196$$

$$x_2 = 1.15 + \frac{3 - 1.15}{\phi} \approx 1.15 + 1.14 = 2.29, \quad f(2.29) \approx (2.29 - 2)^2 + 1 \approx 0.0841 + 1 = 1.0841$$

Since $f(1.86) < f(2.29)$, new interval $[1.15, 2.29]$.

Solved Problem 2: Solution (Continued)

Step 3: Iteration 2

$$x_1 = 3 - \frac{3 - 1.15}{\phi} \approx 3 - 1.14 = 1.86, \quad f(1.86) \approx (1.86 - 2)^2 + 1 \approx 0.0196 + 1 = 1.0196$$

$$x_2 = 1.15 + \frac{3 - 1.15}{\phi} \approx 1.15 + 1.14 = 2.29, \quad f(2.29) \approx (2.29 - 2)^2 + 1 \approx 0.0841 + 1 = 1.0841$$

Since $f(1.86) < f(2.29)$, new interval $[1.15, 2.29]$. **Step 4: Iteration 3**

Solved Problem 2: Solution (Continued)

Step 3: Iteration 2

$$x_1 = 3 - \frac{3 - 1.15}{\phi} \approx 3 - 1.14 = 1.86, \quad f(1.86) \approx (1.86 - 2)^2 + 1 \approx 0.0196 + 1 = 1.0196$$

$$x_2 = 1.15 + \frac{3 - 1.15}{\phi} \approx 1.15 + 1.14 = 2.29, \quad f(2.29) \approx (2.29 - 2)^2 + 1 \approx 0.0841 + 1 = 1.0841$$

Since $f(1.86) < f(2.29)$, new interval $[1.15, 2.29]$. **Step 4: Iteration 3**

$$x_1 = 2.29 - \frac{2.29 - 1.15}{\phi} \approx 2.29 - 0.71 = 1.58, \quad f(1.58) \approx (1.58 - 2)^2 + 1 \approx 0.1764 + 1 = 1.1764$$

$$x_2 = 1.15 + \frac{2.29 - 1.15}{\phi} \approx 1.15 + 0.71 = 1.86 \text{ (reuse)}, \quad f(1.86) \approx 1.0196$$

Solved Problem 2: Solution (Continued)

Step 3: Iteration 2

$$x_1 = 3 - \frac{3 - 1.15}{\phi} \approx 3 - 1.14 = 1.86, \quad f(1.86) \approx (1.86 - 2)^2 + 1 \approx 0.0196 + 1 = 1.0196$$

$$x_2 = 1.15 + \frac{3 - 1.15}{\phi} \approx 1.15 + 1.14 = 2.29, \quad f(2.29) \approx (2.29 - 2)^2 + 1 \approx 0.0841 + 1 = 1.0841$$

Since $f(1.86) < f(2.29)$, new interval $[1.15, 2.29]$. **Step 4: Iteration 3**

$$x_1 = 2.29 - \frac{2.29 - 1.15}{\phi} \approx 2.29 - 0.71 = 1.58, \quad f(1.58) \approx (1.58 - 2)^2 + 1 \approx 0.1764 + 1 = 1.1764$$

$$x_2 = 1.15 + \frac{2.29 - 1.15}{\phi} \approx 1.15 + 0.71 = 1.86 \text{ (reuse)}, \quad f(1.86) \approx 1.0196$$

Since $f(1.58) > f(1.86)$, new interval $[1.58, 2.29]$.

Solved Problem 2: Solution (Continued)

Step 3: Iteration 2

$$x_1 = 3 - \frac{3-1.15}{\phi} \approx 3 - 1.14 = 1.86, \quad f(1.86) \approx (1.86-2)^2+1 \approx 0.0196+1 = 1.0196$$

$$x_2 = 1.15 + \frac{3-1.15}{\phi} \approx 1.15+1.14 = 2.29, \quad f(2.29) \approx (2.29-2)^2+1 \approx 0.0841+1 = 1.0841$$

Since $f(1.86) < f(2.29)$, new interval $[1.15, 2.29]$. **Step 4: Iteration 3**

$$x_1 = 2.29 - \frac{2.29-1.15}{\phi} \approx 2.29 - 0.71 = 1.58, \quad f(1.58) \approx (1.58-2)^2+1 \approx 0.1764+1 = 1.1764$$

$$x_2 = 1.15 + \frac{2.29-1.15}{\phi} \approx 1.15 + 0.71 = 1.86 \text{ (reuse)}, \quad f(1.86) \approx 1.0196$$

Since $f(1.58) > f(1.86)$, new interval $[1.58, 2.29]$. **Convergence**: Interval length $0.71 > 0.5$. Approximate minimum at $x^* = 2$, $f(2) = 1$.

Solved Problem 2: Solution (Continued)

Step 3: Iteration 2

$$x_1 = 3 - \frac{3 - 1.15}{\phi} \approx 3 - 1.14 = 1.86, \quad f(1.86) \approx (1.86-2)^2 + 1 \approx 0.0196 + 1 = 1.0196$$

$$x_2 = 1.15 + \frac{3 - 1.15}{\phi} \approx 1.15 + 1.14 = 2.29, \quad f(2.29) \approx (2.29-2)^2 + 1 \approx 0.0841 + 1 = 1.0841$$

Since $f(1.86) < f(2.29)$, new interval $[1.15, 2.29]$. **Step 4: Iteration 3**

$$x_1 = 2.29 - \frac{2.29 - 1.15}{\phi} \approx 2.29 - 0.71 = 1.58, \quad f(1.58) \approx (1.58-2)^2 + 1 \approx 0.1764 + 1 = 1.1764$$

$$x_2 = 1.15 + \frac{2.29 - 1.15}{\phi} \approx 1.15 + 0.71 = 1.86 \text{ (reuse)}, \quad f(1.86) \approx 1.0196$$

Since $f(1.58) > f(1.86)$, new interval $[1.58, 2.29]$. **Convergence**: Interval length $0.71 > 0.5$. Approximate minimum at $x^* = 2$, $f(2) = 1$. **Verification**: $f'(x) = 2(x - 2) = 0$ at $x = 2$, $f''(x) = 2 > 0$ (minimum).

Solved Problem 3: Golden Section Search

Minimize $f(x) = (x+1)^2$ on $[-2, 2]$ using Golden Section Search.

Solved Problem 3: Golden Section Search

Minimize $f(x) = (x+1)^2$ on $[-2, 2]$ using Golden Section Search. **Step 1: Initialize**

Solved Problem 3: Golden Section Search

Minimize $f(x) = (x+1)^2$ on $[-2, 2]$ using Golden Section Search. **Step 1: Initialize** Interval $[a, b] = [-2, 2]$, $\phi \approx 1.618$, tolerance $\epsilon = 0.5$.

Solved Problem 3: Golden Section Search

Minimize $f(x) = (x+1)^2$ on $[-2, 2]$ using Golden Section Search. **Step 1: Initialize** Interval $[a, b] = [-2, 2]$, $\phi \approx 1.618$, tolerance $\epsilon = 0.5$. **Step 2: Iteration 1**

Solved Problem 3: Golden Section Search

Minimize $f(x) = (x+1)^2$ on $[-2, 2]$ using Golden Section Search. **Step 1: Initialize** Interval $[a, b] = [-2, 2]$, $\phi \approx 1.618$, tolerance $\epsilon = 0.5$. **Step 2: Iteration 1**

$$x_1 = 2 - \frac{2-(-2)}{\phi} \approx 2 - 2.47 = -0.47, \quad f(-0.47) = (-0.47+1)^2 \approx 0.53^2 \approx 0.2809$$

$$x_2 = -2 + \frac{4}{\phi} \approx -2 + 2.47 = 0.47, \quad f(0.47) = (0.47+1)^2 \approx 1.47^2 \approx 2.1609$$

Solved Problem 3: Golden Section Search

Minimize $f(x) = (x+1)^2$ on $[-2, 2]$ using Golden Section Search. **Step 1: Initialize** Interval $[a, b] = [-2, 2]$, $\phi \approx 1.618$, tolerance $\epsilon = 0.5$. **Step 2: Iteration 1**

$$x_1 = 2 - \frac{2-(-2)}{\phi} \approx 2 - 2.47 = -0.47, \quad f(-0.47) = (-0.47+1)^2 \approx 0.53^2 \approx 0.2809$$

$$x_2 = -2 + \frac{4}{\phi} \approx -2 + 2.47 = 0.47, \quad f(0.47) = (0.47+1)^2 \approx 1.47^2 \approx 2.1609$$

Since $f(-0.47) < f(0.47)$, new interval $[-2, 0.47]$.

$$x_2 = -2 + 0.95 \approx -1.05, \quad f(-1.05) \approx 0.0025$$

Since $f(-1.42) > f(-1.05)$, new interval $[-1.42, -0.47]$. **Convergence**: Approximate minimum at $x^* = -1$, $f(-1) = 0$. **Verification**: $f'(x) = 2(x+1) = 0$ at $x = -1$, $f''(x) = 2 > 0$ (minimum).

$$\text{Length} = 2.47,$$

$$\frac{2.47}{\phi \approx 1.53 x_1 = 0.47 - 1.53 \approx -1.06, \quad f(-1.06) = (-1.06+1)^2 \approx (-0.06)^2 = 0.0036}$$

$$x_2 = -2 + 1.53 \approx -0.47 \text{ (reuse)}, \quad f(-0.47) \approx 0.2809$$

Since $f(-1.06) < f(-0.47)$, new interval $[-2, -0.47]$. **Step 4: Iteration 3** Length ≈ 1.53, $\frac{1.53}{\phi} \approx 0.95 x_1 = -0.47 - 0.95 \approx -1.42, \quad f(-1.42) = (-1.42+1)^2 \approx 0.1764$

$$x_2 = -2 + 0.95 \approx -1.05, \quad f(-1.05) \approx 0.0025$$

Since $f(-1.42) > f(-1.05)$, new interval $[-1.42, -0.47]$. **Convergence**: Approximate minimum at $x^* = -1$, $f(-1) = 0$. **Verification**: $f'(x) = 2(x+1) = 0$ at $x = -1$, $f''(x) = 2 > 0$ (minimum).

Since $f(-1.06) < f(-0.47)$, new interval $[-2, -0.47]$. **Step 4: Iteration 3** Length ≈ 1.53, $\frac{1.53}{\phi} \approx 0.95 x_1 = -0.47 - 0.95 \approx -1.42, \quad f(-1.42) = (-1.42 + 1)^2 \approx 0.1764$

$$x_2 = -2 + 0.95 \approx -1.05, \quad f(-1.05) \approx 0.0025$$

Since $f(-1.42) > f(-1.05)$, new interval $[-1.42, -0.47]$. **Convergence**: Approximate minimum at $x^* = -1$, $f(-1) = 0$. **Verification**: $f'(x) = 2(x+1) = 0$ at $x = -1$, $f''(x) = 2 > 0$ (minimum).

Step 4:

Iteration 3 Length ≈ 1.53, $\frac{1.53}{\phi} \approx 0.95 x_1 = -0.47 - 0.95 \approx -1.42, \quad f(-1.42) = (-1.42 + 1)^2 \approx 0.1764$

$$x_2 = -2 + 0.95 \approx -1.05, \quad f(-1.05) \approx 0.0025$$

Since $f(-1.42) > f(-1.05)$, new interval $[-1.42, -0.47]$. **Convergence**: Approximate minimum at $x^* = -1$, $f(-1) = 0$. **Verification**: $f'(x) = 2(x+1) = 0$ at $x = -1$, $f''(x) = 2 > 0$ (minimum).

$$\text{Length} \approx 1.53, \frac{1.53}{\phi} \approx 0.95 x_1 = -0.47 - 0.95 \approx -1.42, \quad f(-1.42) = (-1.42+1)^2 \approx 0.1764$$

$$x_2 = -2 + 0.95 \approx -1.05, \quad f(-1.05) \approx 0.0025$$

Step 3: Iteration 2 Length $= 2.47$,

$2.47 \overline{\phi \approx 1.53 x_1 = 0.47 - 1.53 \approx -1.06, \quad f(-1.06) = (-1.06+1)^2 \approx (-0.06)^2 = 0.0036}$

$$x_2 = -2 + 1.53 \approx -0.47 \text{ (reuse)}, \quad f(-0.47) \approx 0.2809$$

Since $f(-1.06) < f(-0.47)$, new interval $[-2, -0.47]$. **Step 4: Iteration 3** Length ≈ 1.53, $\frac{1.53}{\phi} \approx 0.95 x_1 = -0.47 - 0.95 \approx -1.42, \quad f(-1.42) = (-1.42 + 1)^2 \approx 0.1764$

$$x_2 = -2 + 0.95 \approx -1.05, \quad f(-1.05) \approx 0.0025$$

Since $f(-1.42) > f(-1.05)$, new interval $[-1.42, -0.47]$.

Step 3: Iteration 2 Length $= 2.47$,
$2.47 \frac{}{\phi \approx 1.53 x_1 = 0.47 - 1.53 \approx -1.06, \quad f(-1.06) = (-1.06+1)^2 \approx (-0.06)^2 = 0.0036}$

$$x_2 = -2 + 1.53 \approx -0.47 \text{ (reuse)}, \quad f(-0.47) \approx 0.2809$$

Since $f(-1.06) < f(-0.47)$, new interval $[-2, -0.47]$. **Step 4: Iteration 3** Length ≈ 1.53, $\frac{1.53}{\phi} \approx 0.95 x_1 = -0.47 - 0.95 \approx -1.42, \quad f(-1.42) = (-1.42 + 1)^2 \approx 0.1764$

$$x_2 = -2 + 0.95 \approx -1.05, \quad f(-1.05) \approx 0.0025$$

Since $f(-1.42) > f(-1.05)$, new interval $[-1.42, -0.47]$. **Convergence**: Approximate minimum at $x^* = -1$, $f(-1) = 0$.

Step 3: Iteration 2 Length $= 2.47$,

$2.47 \overline{\phi \approx 1.53 x_1 = 0.47 - 1.53 \approx -1.06, \quad f(-1.06) = (-1.06+1)^2 \approx (-0.06)^2 = 0.0036}$

$$x_2 = -2 + 1.53 \approx -0.47 \text{ (reuse)}, \quad f(-0.47) \approx 0.2809$$

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$$x_2 = -2 + 0.95 \approx -1.05, \quad f(-1.05) \approx 0.0025$$

Since $f(-1.42) > f(-1.05)$, new interval $[-1.42, -0.47]$. **Convergence**: Approximate minimum at $x^* = -1$, $f(-1) = 0$. **Verification**: $f'(x) = 2(x+1) = 0$ at $x = -1$, $f''(x) = 2 > 0$ (minimum).

Step 3: Iteration 2 Length $= 2.47$,

$2.47 \frac{}{\phi \approx 1.53 x_1 = 0.47 - 1.53 \approx -1.06, \quad f(-1.06) = (-1.06+1)^2 \approx (-0.06)^2 = 0.0036}$

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Solved Problem 1: Gradient Method

Minimize $f(w_1, w_2) = w_1^2 + w_2^2 + w_1 w_2$ (neural network loss) using Steepest Descent.

Solved Problem 1: Gradient Method

Minimize $f(w_1, w_2) = w_1^2 + w_2^2 + w_1 w_2$ (neural network loss) using Steepest Descent.
Step 1: Compute Gradient

Solved Problem 1: Gradient Method

Minimize $f(w_1, w_2) = w_1^2 + w_2^2 + w_1 w_2$ (neural network loss) using Steepest Descent.
Step 1: Compute Gradient

$$\nabla f(w_1, w_2) = \begin{bmatrix} 2w_1 + w_2 \\ w_1 + 2w_2 \end{bmatrix}$$

Solved Problem 1: Gradient Method

Minimize $f(w_1, w_2) = w_1^2 + w_2^2 + w_1 w_2$ (neural network loss) using Steepest Descent.
Step 1: Compute Gradient

$$\nabla f(w_1, w_2) = \begin{bmatrix} 2w_1 + w_2 \\ w_1 + 2w_2 \end{bmatrix}$$

Step 2: Initialize

Solved Problem 1: Gradient Method

Minimize $f(w_1, w_2) = w_1^2 + w_2^2 + w_1 w_2$ (neural network loss) using Steepest Descent.
Step 1: Compute Gradient

$$\nabla f(w_1, w_2) = \begin{bmatrix} 2w_1 + w_2 \\ w_1 + 2w_2 \end{bmatrix}$$

Step 2: Initialize Start at $\mathbf{w}_0 = (1, 1)$, step size $\alpha = 0.1$.

Solved Problem 1: Gradient Method

Minimize $f(w_1, w_2) = w_1^2 + w_2^2 + w_1 w_2$ (neural network loss) using Steepest Descent.
Step 1: Compute Gradient

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Step 2: Initialize Start at $\mathbf{w}_0 = (1, 1)$, step size $\alpha = 0.1$. **Step 3: Iterate**

Solved Problem 1: Gradient Method

Minimize $f(w_1, w_2) = w_1^2 + w_2^2 + w_1 w_2$ (neural network loss) using Steepest Descent.
Step 1: Compute Gradient

$$\nabla f(w_1, w_2) = \begin{bmatrix} 2w_1 + w_2 \\ w_1 + 2w_2 \end{bmatrix}$$

Step 2: Initialize Start at $\mathbf{w}_0 = (1, 1)$, step size $\alpha = 0.1$. **Step 3: Iterate**

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \alpha \nabla f(\mathbf{w}_k)$$

Solved Problem 1: Solution (Continued)

Iteration 1:

Solved Problem 1: Solution (Continued)

Iteration 1:

$$\nabla f(1,1) = \begin{bmatrix} 2 \cdot 1 + 1 \\ 1 + 2 \cdot 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.1 \begin{bmatrix} 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 - 0.3 \\ 1 - 0.3 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 0.7 \end{bmatrix}$$

Solved Problem 1: Solution (Continued)

Iteration 1:

$$\nabla f(1,1) = \begin{bmatrix} 2 \cdot 1 + 1 \\ 1 + 2 \cdot 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

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Iteration 2:

Solved Problem 1: Solution (Continued)

Iteration 1:

$$\nabla f(1,1) = \begin{bmatrix} 2 \cdot 1 + 1 \\ 1 + 2 \cdot 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

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Iteration 2:

$$\nabla f(0.7,0.7) = \begin{bmatrix} 2 \cdot 0.7 + 0.7 \\ 0.7 + 2 \cdot 0.7 \end{bmatrix} = \begin{bmatrix} 2.1 \\ 2.1 \end{bmatrix}$$

$$\mathbf{w}_2 = \begin{bmatrix} 0.7 \\ 0.7 \end{bmatrix} - 0.1 \begin{bmatrix} 2.1 \\ 2.1 \end{bmatrix} = \begin{bmatrix} 0.7 - 0.21 \\ 0.7 - 0.21 \end{bmatrix} = \begin{bmatrix} 0.49 \\ 0.49 \end{bmatrix}$$

Solved Problem 1: Solution (Continued)

Iteration 1:

$$\nabla f(1,1) = \begin{bmatrix} 2\cdot 1 + 1 \\ 1 + 2\cdot 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

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Iteration 2:

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Convergence: Iterations approach $\mathbf{w}^* = (0,0)$.

Solved Problem 1: Solution (Continued)

Iteration 1:

$$\nabla f(1,1) = \begin{bmatrix} 2\cdot 1 + 1 \\ 1 + 2\cdot 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

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Convergence: Iterations approach $\mathbf{w}^* = (0,0)$. **Verification**: At $(0,0)$, $\nabla f = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$,

$f(0,0) = 0$.

Solved Problem 1: Solution (Continued)

Iteration 1:

$$\nabla f(1,1) = \begin{bmatrix} 2 \cdot 1 + 1 \\ 1 + 2 \cdot 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

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Convergence: Iterations approach $\mathbf{w}^* = (0,0)$. **Verification**: At $(0,0)$, $\nabla f = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $f(0,0) = 0$. Hessian: $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, det $= 3$, positive definite (minimum).

Solved Problem 1: Solution (Continued)

Iteration 1:

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Iteration 2:

$$\nabla f(0.7,0.7) = \begin{bmatrix} 2 \cdot 0.7 + 0.7 \\ 0.7 + 2 \cdot 0.7 \end{bmatrix} = \begin{bmatrix} 2.1 \\ 2.1 \end{bmatrix}$$

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Solved Problem 1: Solution (Continued)

Iteration 1:

$$\nabla f(1,1) = \begin{bmatrix} 2 \cdot 1 + 1 \\ 1 + 2 \cdot 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

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Iteration 2:

$$\nabla f(0.7,0.7) = \begin{bmatrix} 2 \cdot 0.7 + 0.7 \\ 0.7 + 2 \cdot 0.7 \end{bmatrix} = \begin{bmatrix} 2.1 \\ 2.1 \end{bmatrix}$$

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Solved Problem 2: Gradient Method

Minimize $f(x, y) = (x - 1)^2 + (y - 2)^2$ using Steepest Descent.

Solved Problem 2: Gradient Method

Minimize $f(x, y) = (x-1)^2 + (y-2)^2$ using Steepest Descent. **Step 1: Compute Gradient**

Solved Problem 2: Gradient Method

Minimize $f(x, y) = (x-1)^2 + (y-2)^2$ using Steepest Descent. **Step 1: Compute Gradient**

$$\nabla f(x, y) = \begin{bmatrix} 2(x-1) \\ 2(y-2) \end{bmatrix}$$

Solved Problem 2: Gradient Method

Minimize $f(x,y) = (x-1)^2 + (y-2)^2$ using Steepest Descent. **Step 1: Compute Gradient**

$$\nabla f(x,y) = \begin{bmatrix} 2(x-1) \\ 2(y-2) \end{bmatrix}$$

Step 2: Initialize

Solved Problem 2: Gradient Method

Minimize $f(x, y) = (x-1)^2 + (y-2)^2$ using Steepest Descent. **Step 1: Compute Gradient**

$$\nabla f(x, y) = \begin{bmatrix} 2(x-1) \\ 2(y-2) \end{bmatrix}$$

Step 2: Initialize Start at $\mathbf{w}_0 = (0, 0)$, step size $\alpha = 0.1$.

Solved Problem 2: Gradient Method

Minimize $f(x, y) = (x-1)^2 + (y-2)^2$ using Steepest Descent. **Step 1: Compute Gradient**

$$\nabla f(x, y) = \begin{bmatrix} 2(x-1) \\ 2(y-2) \end{bmatrix}$$

Step 2: Initialize Start at $\mathbf{w}_0 = (0, 0)$, step size $\alpha = 0.1$. **Step 3: Iterate**

Solved Problem 2: Gradient Method

Minimize $f(x, y) = (x-1)^2 + (y-2)^2$ using Steepest Descent. **Step 1: Compute Gradient**

$$\nabla f(x, y) = \begin{bmatrix} 2(x-1) \\ 2(y-2) \end{bmatrix}$$

Step 2: Initialize Start at $\mathbf{w}_0 = (0, 0)$, step size $\alpha = 0.1$. **Step 3: Iterate**

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \alpha \nabla f(\mathbf{w}_k)$$

Solved Problem 2: Solution (Continued)

Iteration 1:

Solved Problem 2: Solution (Continued)

Iteration 1:

$$\nabla f(0,0) = \begin{bmatrix} 2(0-1) \\ 2(0-2) \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} -2 \\ -4 \end{bmatrix} = \begin{bmatrix} 0 + 0.2 \\ 0 + 0.4 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

Solved Problem 2: Solution (Continued)

Iteration 1:

$$\nabla f(0,0) = \begin{bmatrix} 2(0-1) \\ 2(0-2) \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} -2 \\ -4 \end{bmatrix} = \begin{bmatrix} 0+0.2 \\ 0+0.4 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

Iteration 2:

Solved Problem 2: Solution (Continued)

Iteration 1:

$$\nabla f(0,0) = \begin{bmatrix} 2(0-1) \\ 2(0-2) \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} -2 \\ -4 \end{bmatrix} = \begin{bmatrix} 0+0.2 \\ 0+0.4 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

Iteration 2:

$$\nabla f(0.2,0.4) = \begin{bmatrix} 2(0.2-1) \\ 2(0.4-2) \end{bmatrix} = \begin{bmatrix} -1.6 \\ -3.2 \end{bmatrix}$$

$$\mathbf{w}_2 = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix} - 0.1 \begin{bmatrix} -1.6 \\ -3.2 \end{bmatrix} = \begin{bmatrix} 0.2+0.16 \\ 0.4+0.32 \end{bmatrix} = \begin{bmatrix} 0.36 \\ 0.72 \end{bmatrix}$$

Solved Problem 2: Solution (Continued)

Iteration 1:

$$\nabla f(0,0) = \begin{bmatrix} 2(0-1) \\ 2(0-2) \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} -2 \\ -4 \end{bmatrix} = \begin{bmatrix} 0+0.2 \\ 0+0.4 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

Iteration 2:

$$\nabla f(0.2,0.4) = \begin{bmatrix} 2(0.2-1) \\ 2(0.4-2) \end{bmatrix} = \begin{bmatrix} -1.6 \\ -3.2 \end{bmatrix}$$

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Convergence: Iterations approach $\mathbf{w}^* = (1,2)$.

Solved Problem 2: Solution (Continued)

Iteration 1:

$$\nabla f(0,0) = \begin{bmatrix} 2(0-1) \\ 2(0-2) \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} -2 \\ -4 \end{bmatrix} = \begin{bmatrix} 0+0.2 \\ 0+0.4 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

Iteration 2:

$$\nabla f(0.2,0.4) = \begin{bmatrix} 2(0.2-1) \\ 2(0.4-2) \end{bmatrix} = \begin{bmatrix} -1.6 \\ -3.2 \end{bmatrix}$$

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Convergence: Iterations approach $\mathbf{w}^* = (1,2)$. **Verification**: At $(1,2)$, $\nabla f = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$,

$f(1,2) = 0$.

Solved Problem 2: Solution (Continued)

Iteration 1:

$$\nabla f(0,0) = \begin{bmatrix} 2(0-1) \\ 2(0-2) \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} -2 \\ -4 \end{bmatrix} = \begin{bmatrix} 0+0.2 \\ 0+0.4 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

Iteration 2:

$$\nabla f(0.2,0.4) = \begin{bmatrix} 2(0.2-1) \\ 2(0.4-2) \end{bmatrix} = \begin{bmatrix} -1.6 \\ -3.2 \end{bmatrix}$$

$$\mathbf{w}_2 = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix} - 0.1 \begin{bmatrix} -1.6 \\ -3.2 \end{bmatrix} = \begin{bmatrix} 0.2+0.16 \\ 0.4+0.32 \end{bmatrix} = \begin{bmatrix} 0.36 \\ 0.72 \end{bmatrix}$$

Convergence: Iterations approach $\mathbf{w}^* = (1,2)$. **Verification**: At $(1,2)$, $\nabla f = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $f(1,2) = 0$. Hessian: $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, positive definite (minimum).

Solved Problem 2: Solution (Continued)

Iteration 1:

$$\nabla f(0,0) = \begin{bmatrix} 2(0-1) \\ 2(0-2) \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} -2 \\ -4 \end{bmatrix} = \begin{bmatrix} 0+0.2 \\ 0+0.4 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

Iteration 2:

$$\nabla f(0.2,0.4) = \begin{bmatrix} 2(0.2-1) \\ 2(0.4-2) \end{bmatrix} = \begin{bmatrix} -1.6 \\ -3.2 \end{bmatrix}$$

$$\mathbf{w}_2 = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix} - 0.1 \begin{bmatrix} -1.6 \\ -3.2 \end{bmatrix} = \begin{bmatrix} 0.2+0.16 \\ 0.4+0.32 \end{bmatrix} = \begin{bmatrix} 0.36 \\ 0.72 \end{bmatrix}$$

Convergence: Iterations approach $\mathbf{w}^* = (1,2)$. **Verification**: At $(1,2)$, $\nabla f = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $f(1,2) = 0$. Hessian: $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, positive definite (minimum). **Solution**: Minimum at $(x,y) = (1,2)$, $f = 0$.

Solved Problem 2: Solution (Continued)

Iteration 1:

$$\nabla f(0,0) = \begin{bmatrix} 2(0-1) \\ 2(0-2) \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} -2 \\ -4 \end{bmatrix} = \begin{bmatrix} 0+0.2 \\ 0+0.4 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

Iteration 2:

$$\nabla f(0.2,0.4) = \begin{bmatrix} 2(0.2-1) \\ 2(0.4-2) \end{bmatrix} = \begin{bmatrix} -1.6 \\ -3.2 \end{bmatrix}$$

$$\mathbf{w}_2 = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix} - 0.1 \begin{bmatrix} -1.6 \\ -3.2 \end{bmatrix} = \begin{bmatrix} 0.2+0.16 \\ 0.4+0.32 \end{bmatrix} = \begin{bmatrix} 0.36 \\ 0.72 \end{bmatrix}$$

Convergence: Iterations approach $\mathbf{w}^* = (1,2)$. **Verification**: At $(1,2)$, $\nabla f = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $f(1,2) = 0$. Hessian: $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, positive definite (minimum). **Solution**: Minimum at $(x,y) = (1,2)$, $f = 0$. **Interpretation**: Converges to optimal point efficiently with appropriate α.

Solved Problem 3: Gradient Method

Minimize $f(x, y) = x^2 + 2y^2 - 2x - 4y$ using Steepest Descent.

Solved Problem 3: Gradient Method

Minimize $f(x, y) = x^2 + 2y^2 - 2x - 4y$ using Steepest Descent. **Step 1: Compute Gradient**

Solved Problem 3: Gradient Method

Minimize $f(x, y) = x^2 + 2y^2 - 2x - 4y$ using Steepest Descent. **Step 1: Compute Gradient**

$$\nabla f(x, y) = \begin{bmatrix} 2x - 2 \\ 4y - 4 \end{bmatrix}$$

Solved Problem 3: Gradient Method

Minimize $f(x, y) = x^2 + 2y^2 - 2x - 4y$ using Steepest Descent. **Step 1: Compute Gradient**

$$\nabla f(x, y) = \begin{bmatrix} 2x - 2 \\ 4y - 4 \end{bmatrix}$$

Step 2: Initialize

Solved Problem 3: Gradient Method

Minimize $f(x, y) = x^2 + 2y^2 - 2x - 4y$ using Steepest Descent. **Step 1: Compute Gradient**

$$\nabla f(x, y) = \begin{bmatrix} 2x - 2 \\ 4y - 4 \end{bmatrix}$$

Step 2: Initialize Start at $(3, 0)$, step size $\alpha = 0.1$.

Solved Problem 3: Gradient Method

Minimize $f(x, y) = x^2 + 2y^2 - 2x - 4y$ using Steepest Descent. **Step 1: Compute Gradient**

$$\nabla f(x, y) = \begin{bmatrix} 2x - 2 \\ 4y - 4 \end{bmatrix}$$

Step 2: Initialize Start at $(3, 0)$, step size $\alpha = 0.1$. **Step 3: Iterate**

Solved Problem 3: Gradient Method

Minimize $f(x, y) = x^2 + 2y^2 - 2x - 4y$ using Steepest Descent. **Step 1: Compute Gradient**

$$\nabla f(x, y) = \begin{bmatrix} 2x - 2 \\ 4y - 4 \end{bmatrix}$$

Step 2: Initialize Start at $(3, 0)$, step size $\alpha = 0.1$. **Step 3: Iterate**

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \alpha \nabla f(\mathbf{w}_k)$$

Solved Problem 3: Solution (Continued)

Iteration 1:

Solved Problem 3: Solution (Continued)

Iteration 1:

$$\nabla f(3,0) = \begin{bmatrix} 2 \cdot 3 - 2 \\ 4 \cdot 0 - 4 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} 4 \\ -4 \end{bmatrix} = \begin{bmatrix} 3 - 0.4 \\ 0 + 0.4 \end{bmatrix} = \begin{bmatrix} 2.6 \\ 0.4 \end{bmatrix}$$

Solved Problem 3: Solution (Continued)

Iteration 1:

$$\nabla f(3,0) = \begin{bmatrix} 2 \cdot 3 - 2 \\ 4 \cdot 0 - 4 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \end{bmatrix}$$

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Iteration 2:

Solved Problem 3: Solution (Continued)

Iteration 1:

$$\nabla f(3,0) = \begin{bmatrix} 2 \cdot 3 - 2 \\ 4 \cdot 0 - 4 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} 4 \\ -4 \end{bmatrix} = \begin{bmatrix} 3 - 0.4 \\ 0 + 0.4 \end{bmatrix} = \begin{bmatrix} 2.6 \\ 0.4 \end{bmatrix}$$

Iteration 2:

$$\nabla f(2.6, 0.4) = \begin{bmatrix} 2 \cdot 2.6 - 2 \\ 4 \cdot 0.4 - 4 \end{bmatrix} = \begin{bmatrix} 3.2 \\ -2.4 \end{bmatrix}$$

$$\mathbf{w}_2 = \begin{bmatrix} 2.6 \\ 0.4 \end{bmatrix} - 0.1 \begin{bmatrix} 3.2 \\ -2.4 \end{bmatrix} = \begin{bmatrix} 2.6 - 0.32 \\ 0.4 + 0.24 \end{bmatrix} = \begin{bmatrix} 2.28 \\ 0.64 \end{bmatrix}$$

Solved Problem 3: Solution (Continued)

Iteration 1:

$$\nabla f(3,0) = \begin{bmatrix} 2 \cdot 3 - 2 \\ 4 \cdot 0 - 4 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \end{bmatrix}$$

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Iteration 2:

$$\nabla f(2.6,0.4) = \begin{bmatrix} 2 \cdot 2.6 - 2 \\ 4 \cdot 0.4 - 4 \end{bmatrix} = \begin{bmatrix} 3.2 \\ -2.4 \end{bmatrix}$$

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Convergence: Iterations approach $\mathbf{w}^* = (1,1)$.

Solved Problem 3: Solution (Continued)

Iteration 1:

$$\nabla f(3,0) = \begin{bmatrix} 2\cdot 3-2 \\ 4\cdot 0-4 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \end{bmatrix}$$

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Convergence: Iterations approach $\mathbf{w}^* = (1,1)$. **Verification**: At $(1,1)$, $\nabla f = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$,

$f(1,1) = 1+2-2-4 = -3$.

Solved Problem 3: Solution (Continued)

Iteration 1:

$$\nabla f(3,0) = \begin{bmatrix} 2 \cdot 3 - 2 \\ 4 \cdot 0 - 4 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \end{bmatrix}$$

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Convergence: Iterations approach $\mathbf{w}^* = (1,1)$. **Verification**: At $(1,1)$, $\nabla f = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $f(1,1) = 1 + 2 - 2 - 4 = -3$. Hessian: $\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$, positive definite (minimum).

Solved Problem 3: Solution (Continued)

Iteration 1:

$$\nabla f(3,0) = \begin{bmatrix} 2 \cdot 3 - 2 \\ 4 \cdot 0 - 4 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \end{bmatrix}$$

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Convergence: Iterations approach $\mathbf{w}^* = (1,1)$. **Verification**: At $(1,1)$, $\nabla f = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $f(1,1) = 1 + 2 - 2 - 4 = -3$. Hessian: $\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$, positive definite (minimum).

Solution: Minimum at $(x,y) = (1,1)$, $f = -3$.

Solved Problem 3: Solution (Continued)

Iteration 1:

$$\nabla f(3,0) = \begin{bmatrix} 2 \cdot 3 - 2 \\ 4 \cdot 0 - 4 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \end{bmatrix}$$

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Iteration 2:

$$\nabla f(2.6,0.4) = \begin{bmatrix} 2 \cdot 2.6 - 2 \\ 4 \cdot 0.4 - 4 \end{bmatrix} = \begin{bmatrix} 3.2 \\ -2.4 \end{bmatrix}$$

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Convergence: Iterations approach $\mathbf{w}^* = (1,1)$. **Verification**: At $(1,1)$, $\nabla f = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $f(1,1) = 1 + 2 - 2 - 4 = -3$. Hessian: $\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$, positive definite (minimum).
Solution: Minimum at $(x,y) = (1,1)$, $f = -3$. **Interpretation**: Optimizes quadratic loss function.

Application: Resource Allocation

Maximize $f(x, y) = 30x + 50y - 2x^2 - 2y$ using Gradient Method (Steepest Ascent).

- Perform two iterations starting at $(0, 0)$, $\alpha = 0.1$.
- Interpret the solution in the context of resource allocation (e.g., labor and capital).
- Discuss the effect of different step sizes.

Exercise 1

Minimize $f(x) = x^2 + 4x + 5$ on $[-2, 2]$ using Dichotomous Search.

Perform two iterations with $\delta = 0.1$.

Estimate the interval of uncertainty.

Verify the approximate minimum.

Exercise 2

Minimize $f(x, y) = x^2 + 2y^2$ using Steepest Descent.

- Start at $(2, 2)$, use $\alpha = 0.05$, perform two iterations.
- Check convergence toward the minimum.
- Discuss the impact of step size.

Assignment 1 (Individual)

Minimize $f(x) = x^4 - 4x^2 + 2$ on $[-3, 3]$ using Golden Section Search.

Perform three iterations with $\epsilon = 0.5$.

Submit a report including:

Iteration steps and interval reduction.
Verification of the minimum.
Discussion of unimodal properties.

Assignment 2 (Team Work)

Minimize $f(w_1, w_2) = w_1^2 + 2w_2^2 + w_1 w_2$ using Steepest Descent.

- Form a team of 3–4 students.
- Perform three iterations starting at $(1, 1)$, $\alpha = 0.1$.
- Prepare a presentation including:
  - Iteration steps and convergence analysis.
  - Interpretation in neural network training.
  - Effect of varying step size.
- Deadline: One week from today.

References

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M. S. Bazaraa, H. J. Sherali, C. M. Shetty, *Nonlinear Programming: Theory and Algorithms*, 3rd ed., Wiley, 2013.

NPTEL: Prof. S. S. Rao,
https://onlinecourses.nptel.ac.in/noc20_me46/preview

Thank You

Questions?